

Alidade wire harnessing: controller box										
Reference ELMCOOverview.vsd										
24-Mar-06										
AWC1 Power from relay board to power mgmt board										
	A	Description	B	Comment	Color	Size	Twist	Length		
Wires	1	100 VAC "L"			White	18		15"		
	2	Return "N"			Black	18		15"		
	3	Ground			Green	18		15"		
Ends		A IEC-320 0.25 quick disconnect F insulated								
		B spade fork term #6, unins bent heat shrink								
AWC2 25-wire cable from alidade relay board										
	A	Description	B,C,D	Comment	Color	Size	Twist	Length		
Wires/pins	1	brakes +24V	C1	power mgmt board	brown	all wires #22 AWG	Z	7.5"		
	2	Ground	C2		black		Z	7.5"		
	3	24V aux	D1		purple		Y	7.5"		
	4	Ground	D2		black		Y	7.5"		
	5	+5V	B19	microcontroller board	red		X	7.5"		
	6	Ground	B15		black		X	7.5"		
	7	float in	B6		green			7.5"		
	8	T 2 input	B9		blue			7.5"		
	9	T 4 input	B10		yellow			7.5"		
	10	Az brake sense	B13		white			7.5"		
	11	Latch	B7		orange			7.5"		
	12	Pot 1 (GND)	B18		black		W	7.5"		
	13	Pot wiper	B16		green			7.5"		
	14	brakes +24V	C1	power mgmt board	brown		Z	7.5"		
	15	Ground	C2		black		Z	7.5"		
	16	24V aux	D1		purple		Y	7.5"		
	17	Ground	D2		black		Y	7.5"		
	18	+5V	B20	microcontroller board	red		X	7.5"		
	19	Ground	B3		black		X	7.5"		
	20	T 1 input	B11		green			7.5"		
	21	T 3 input	B12		blue			7.5"		
	22	electronics reset (on/off)	B5		yellow			7.5"		
	23	EI brake sense	B8		white			7.5"		
	24	Unlatch (drives on/off)	B14		grey			7.5"		
	25	Pot 2 (+5V)	B17		red		W	7.5"		
Ends		A 56-126-004	filtered D25 Socket on top of control b			connects to AWE7B				
		B WM2527-ND/WM62510-ND	2x10 100mil ucntlr board			(22-55-2201/16-02-0102)				
		C 770849-2/770522-1	power mgmt board		relay/brakes		JameCo 236321			
		D 770849-2/770522-1	power mgmt board		HV aux		JameCo 236321			
B:										
	1: N/C	3: GND	5: RST	7: LTCH	9: T2	11: T1	13: ABS	15: GND	17: P5V	19: +5V
	2: N/C	4: N/C	6: FLT	8: EBS	10: T4	12: T3	14: ULTC	16: WPR	18: GND	20: +5V
AWC3 encoder for azimuth (P/N: R176H-03600Q-5L10-AT40SP-24MN)										
	A	Description	B	Comment	Color	Size	Twist	Length		
Wires/pins	1	A	3		yellow	all wires #24 AWG	Z	5.75		
	2	B	5		green		Y	5.75		
	3	IND (Z)	7		blue		X	5.75		
	4	N/C								
	5	+5V	9		red		W	5.75		
	6	/A	4		brown		Z	5.75		
	7	/B	6		orange		Y	5.75		
	8	/IND (Z)	8		white		X	5.75		
	9	COMMON	10		black		W	5.75		
Ends		A 56-106-010	filtered Dsub 9 socket on control box			connects to AWE12A				
		B WM2522-ND/WM62510-ND	2x5 0.100" on ucntlr card			(22-55-2101/16-02-0102)				
	1: NC	3: A	5: B	7: Z(IND)	9: +5V					
	2: NC	4: /A	6: /B	8: /Z(IND)	10: GND		in-line w/ battery			

AWC4		encoder for elevation (P/N: 9220S03600D5L10D99SP04EA)						
Wires/pins	A	Description	B	Comment	Color	Size	Twist	Length
	1	A	3		yellow	all wires #24 AWG	Z	5.75
	2	B	5		green		Y	5.75
	3	IND (Z)	7		blue		X	5.75
	4	N/C						
	5	+5V	9		red		W	5.75
	6	/A	4		brown		Z	5.75
	7	/B	6		orange		Y	5.75
	8	/IND (Z)	8		white		X	5.75
	9	COMMON	10		black		W	5.75
Ends	A 56-106-010		D9S				connects to AWE13A	
	B WM2522-ND/WM62510-ND		2x5 0.100" on ucntlr card			(22-55-2101/16-02-0102)		
	1: NC	3: A	5: B	7: Z(IND)	9: +5V			
	2: NC	4: /A	6: /B	8: /Z(IND)	10: GND			
AWC5 battery								
Wires/pins	A	Description	B	Comment	Color	Size	Twist	Length
	1	+12V	1	use 1-#18	red	18		4
	2	+12V	1	or 3-#22	red	same wire		
	3	N/C						
	4	GND	2	use 1-#18	black	18		4
	5	GND	2	or 3-#22	black	same wire		
	6	+12V	1		red	same wire		
	7	N/C						
	8	N/C						
	9	GND	2		black	same wire		
Ends	A 56-101-010		filtered Dsub 9 pin on control box			connects to AWE14A		
	B 770849-2/770522-1							
AWC6								
standard CAT5 cable		RJ-45						12
AWC7 9V to media converter from power mgmt board								
Wires	A	Description	B	Comment	Color	Size	Twist	Length
	1	+9V	1		black/white	20		13
	2	Common	2		black	20		13
Ends	A round							
	B 770849-2/770522-1		This connector numbering is opposite to 0.100" connectors					
AWC8 24 V from power supply to power mgmt board								
Wires	A	Description	B	Comment	Color	Size	Twist	Length
	1	+24V	1		red	18		8.25
	2	Common	2		black	18		8.25
Ends	A spade fork #6x18							
	B 770849-2/770522-1							
AWC9 5V to SBC								
Wires	A	Description	B	Comment	Color	Size	Twist	Length
	1	+5V	1		red	18		7.5
	2	Common	2		black	18		7.5
Ends	A 15-24-4048							
	B 770849-2/770522-1							
AWC10 12 VDC from Power Management to uCntlr board								
Wires	A	Description	B	Comment	Color	Size	Twist	Length
	1	+12V	1		red	18		4.5
	2	Common	2		black	18		4.5
Ends	A 770849-2/770522-1							
	B 770849-2/770522-1			0.156" pitch				

AWC11 SBC reset (goes between the uCntrlr board and SBC)								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	power	1		red	24	Z	9
	2	ground	2		black	24	Z	9
	3	reset	3		green	24	Z	9
	4	N/C	4	pin 1 on inboard side				
	5	N/C	5					
Ends	A	WM2803-ND/WM62510-ND		1x5 100mil connector on ucntrlr card		(50-57-9005/16-02-0102)		
	B	WM2803-ND/WM62510-ND		1x5 100mil connector on SBC		(50-57-9005/16-02-0102)		
AWC12 electronics on/off (between uCntrlr board and power management board)								
	A	Description	B	Comment	Color	Size	Twist	Length
Wires	1	electronics on/off	1		green	24	Z	6.5
	2	ground	2		black	24	Z	6.5
	3	battery voltage /3	3		yellow	24	Z	6.5
Ends	A	50-57-9003/16-02-0102						
	B	50-57-9003/16-02-0102						
AWC13 short cat 5 crossover cable between SBC (second LAN) and uCntrlr board								
crossover CAT5 cable								6"